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Individual Bank risk

Market risk (Whole Bank sector Risk)

$CAPM = R_f + \beta(MRP)$

$= 10\% + 1.1(12\% - 10\%)$

Stock valuation

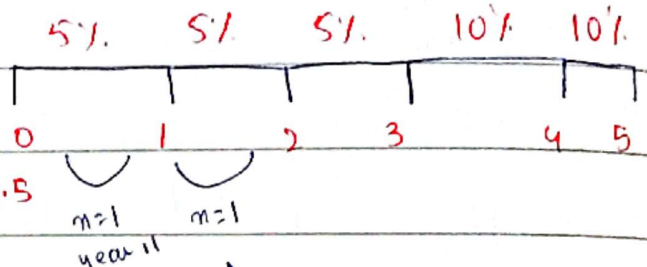
(7-1)

$D_0 = 1.50$

$g = 5\% - 0.05$ for 3 year

$g_1 = 10\% - 0$ for 4 year

$D_1, D_2, D_3, D_4, D_5 = ?$



$D_0 = 1$

Future value formula

$D_1 = D_0(1+g)^n$

$D_1 = 1.5(1+0.05)^1 = 1.575 \rightarrow D_1$

$D_2 = D_1(1+g)^n$

$D_2 = 1.575(1+0.05)^1 = 1.65 \rightarrow D_2$

$D_3 = D_2(1+g)^n$

$D_3 = 1.65(1+0.05)^1 = 1.73 \rightarrow D_3$

$D_4 = 1.73\ D_3(1+g)^n$

$D_4 = 1.73(1+0.10)^1 = 1.903$

$D_5 = D_4(1+g)^n$

$D_5 = 1.903(1+0.10)^1 = 2.09$

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If growth rate is constant then we can apply this formula

$$P_0 = \frac{D_1}{K_s - g}$$

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and if we have non-constant growth rate then we cannot apply this formula.

(7-2)

$D_1 = 1.50\$$

$K_s = 15\% \rightarrow 0.15$

$g = 7\% \rightarrow 0.07$

$$P_0 = \frac{D_1}{K_s - g} \Rightarrow \frac{1.5}{0.15 - 0.07} \Rightarrow \boxed{18.75 = P_0}$$

(7-3)

$P_0 = 20$

$D_0 = 1$

$g = 10\%$

$K_s = ?$

$$P_0 = \frac{D_1}{K_s - g} \Rightarrow \frac{D_0(1+g)}{K_s - g} \Rightarrow \frac{1(1+0.10)}{K_s - 0.10} = 20 \quad (P_0)$$

$20(K_s - 0.10) = 1.1$

$20K_s - 2 = 1.1 \Rightarrow 20K_s = 1.1 + 2$

$20K_s = 3.1 \Rightarrow K_s = 3.1/20 = \boxed{K_s = 15.5\%}$

$$P_1 = \frac{D_1(1+g)}{K_s - g} = \frac{1.1 \times (1+0.10)}{0.155 - 0.10} = \frac{1.21}{0.055} = 22$$

$P = \frac{D}{P}$

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(7-4)

* Required Return = $\frac{\text{Dividend per share}}{\text{Price per share}}$

$D_1 = 5$
$P_0 = 50$

Fixed growth rate

Stocks
Common | Preffered

$P_0 = D_1/(K_s - g)$

$P_0 = \frac{D_0}{K_s} \Rightarrow K_s = \frac{D_1}{P_0} = > \frac{5}{50} = \frac{1}{10} = 10\%$

Cost of preferred stock = $\frac{\text{Prefered stock dividend}}{\text{current price of Pefer stock}}$

(7.5) $D_0 = 2\$$

$B = 1.2$
$R_{fR} = 7.5\%$ } K_s
$MRP = 4\%$

Calcular the price at year 2 bc after this growth is constant

20% 20% 7%
$D_0 = 2\$$

$g = 20\% \rightarrow 2$ year , $7\% \rightarrow$ after 2 year

$K_s = CAPM = R_{rf} + B(MRP)$

$K_s = 7.5 + 1.2(4\%) = 12.3\%$

growth is changing bc non-constant growth rate

$P_0 = \frac{D_0(1+g)}{K_s - g}$

$D_1 = D_0(1+g)^n = 2(1+0.20)^1 = 2.4$
$D_2 = D_1(1+g)^n = 2.4(1+1.0) = 2.8$
$D_3 = D_2(1+g)^n = 2.8(1+0.07) = 3.081$

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use any one formula.

$$P_2 = \frac{D_3}{K_s - g} \rightarrow \frac{D_2(1+g)}{K_s - g}$$

$$P_2 = \frac{3.081}{0.123 - 0.07}$$

$P_2 = \$58$ · Price at year 2 → but we have to calculate at zero year. P_0.

Present value = $\frac{\text{Future value}}{(1+i)^n}$ (+ K_s also)

$$P_0 = \frac{2.4}{(1+0.123)^1} + \frac{2.88 + 58}{(1+0.123)^2} \Rightarrow 2.13 + 48.27$$

$\Rightarrow \$50.40$

Price at year zero

(7-6)

$P_0 = 80\$$

$D_1 = 4\$$, g - constant.

$K_s = 14\% \rightarrow 0.14$

$g = ?$

Sol

$$P_0 = \frac{D_1}{K_s - g} \Rightarrow 80 = \frac{4}{0.14 - g}$$

$$80(0.14 - g) = 4$$

$$11.2 - 80g = 4$$

$$11.2 - 4 = 80g$$

$$7.2/8 = g \rightarrow g = 0.090 \rightarrow 9\%$$

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(7-7)

$D_1 = 2\$$ $\beta = 0.9$

$g = ?$ (constant rate) $R_{rf} = 5.6\%$

$P_0 = 25\$$ MRP = 6%.

$P_3 = ?$

(1) For Ks: CAPM $\Rightarrow$ $K_s = R_{rf} + \beta(MRP)$

$= 5.6\% + 0.9(6\%) \Rightarrow$ **$11\% = K_s$**

For g:

$$P_0 = \frac{D_1}{K_s - g} \Rightarrow 25 = \frac{2}{0.11 - g} \Rightarrow 25(0.11 - g) = 2$$

$2.75 - 2 = 25g$

$0.75/25 = g$

$g = 0.03 \Rightarrow 3\%$

For P_3:

$$P_3 = \frac{D_4}{K_s - g}$$

$D_2 = D_1(1+g)^n \rightarrow 2(1+0.03)^1 \rightarrow 2.06$

$D_3 = D_2(1+g)^n \rightarrow 2.06(1+0.03)^1 \rightarrow 2.12$

$D_4 = D_3(1+g)^n \rightarrow 2.12(1+0.03)^1 \rightarrow 2.18$

$$P_3 = \frac{D_4}{K_s - g} \Rightarrow \frac{2.18}{0.11 - 0.03} \Rightarrow \frac{2.18}{0.08}$$

$P_3 = 27.25\$$

Finance

Organization → Share (cost)

Individual → Share (profit)

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(7-8) Dividend = 8 %.

Price => $P_1 = 60$ (a), $P_2 = 80$ (b), $P_3 = 100$ (c), $P_4 = 140$ (d)

Rate of return = ?

Rate of return = Dividend / Price

(a) $r = D/P \Rightarrow 8/60 = 0.13 \Rightarrow 13\%$

(b) $r = D/P \Rightarrow 8/80 = 0.10 \Rightarrow 10\%$

(c) $r = D/P \Rightarrow 8/100 = 0.08 \Rightarrow 8\%$

(d) $r = D/P \Rightarrow 8/140 = 0.05 \Rightarrow 5\%$

(7-9) $D_0 = 5\$$

$K_s = 15\% \rightarrow 0.15$

$g = -4\% \rightarrow -0.04$ (growth declining)

$P = ?$

$$P_0 = \frac{D_1}{K_s - g} \Rightarrow \frac{D_0(1+g)^n}{K_s - g}$$

$$P_0 \Rightarrow \frac{5(1+(-0.04)^1}{0.15-(-0.04)} \Rightarrow \frac{4.8}{0.19} \Rightarrow 25.26 = P_0$$

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(7*10) (A) ; Stock C | Stock D

$\beta = 0.4$ | $\beta = -0.5$

$R_{rf} = 9\% \rightarrow 0.09$

$R_m = 13\% \rightarrow 0.13$

[$R_{rf} = (r^* + IP)$
$MRP = (R_m - R_{rf})$

CAPM $\Rightarrow K_s = R_{rf} + \beta(MRP)$

$$\boxed{K_s = R_{rf} + \beta(R_m - R_{rf})}$$

Rate of return on Stock C:

$K_s = 0.09 + 0.4(0.13 - 0.09)$

$= 0.106 \Rightarrow \boxed{10.6\%}$

Rate of return on Stock D:

$K_s = 0.09 + [(-0.5)(0.13 - 0.09)]$

$= 0.07 \Rightarrow \boxed{7\%}$

(B) $P_0 = 25\$, \quad D_1 = 1.50\$$

$g = 4\% \rightarrow 0.04$

$K_s = ?$

$$P_0 = \frac{D_1}{K_s - g} \Rightarrow 25 = \frac{1.50}{K_s - 0.04}$$

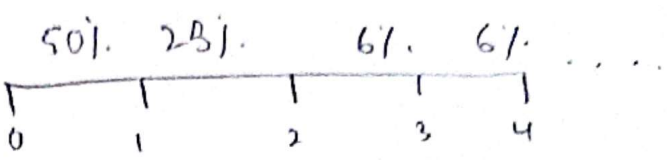
$25(K_s - 0.04) = 1.50$

$25K_s - 1 = 1.50$

$25K_s = 1.50 + 1$

$K_s = 2.5/25 \Rightarrow \boxed{K_s = 10\%}$

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$D_0 = 1\$$

$P_0 = ?$

$P_2 = ?$

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(7-11)

∴ Return on Equity = Div + Capital Gain

$K_s = 6\% + 7\% = 13\%$

$P_0 = ?$, $i/K_S = 6\% + 7\% = 13\%$

$D_0 = 1\$$, $g_1 = 50\%$, $g_2 = 25\%$, $g_3 = 6\%$

$D_1 = D_0(1+g)^n \rightarrow 1(1+0.5)^1 = 1.5$

$D_2 = D_1(1+g)^n \rightarrow 1.5(1+0.25)^1 = 1.87$

$D_3 = D_2(1+g)^n \rightarrow 1.87(1+0.06)^1 = 1.98$

Calculate value at P_2 bcz growth rate constant from this year.

$$P_2 = \frac{D_3}{K_S - g} \Rightarrow \frac{1.98}{0.13 - 0.06} \Rightarrow \boxed{28.28}$$

Calculate value at Present (P_0) time.

$$P_0 = \frac{D_1}{(1+i)^n} + \frac{D_2}{(1+i)^n} + \frac{P_2}{(1+i)^n}$$

$$= \frac{1.5}{(1+0.13)^1} + \frac{1.87}{(1+0.13)^2} + \frac{28.28}{(1+0.13)^2}$$

$$= 1.32 + 1.46 + 22.14$$

$$\boxed{P_0 = 24.92\$}$$

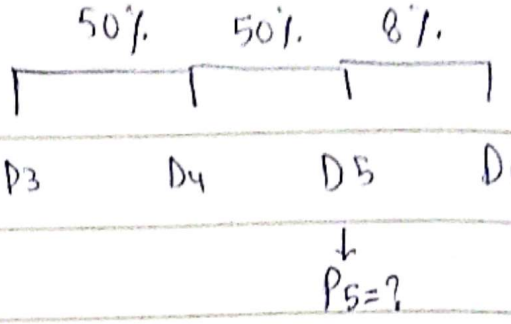
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(7-12) $D_3 = 1\$$

$K_s = 15\% \rightarrow 0.15$

$P_0 = ?$



Calculate the value of stock at year 5 (Constant growth rate)

$$P_5 = \frac{D_6}{K_s - g} = \frac{2.43}{0.15 - 0.08} \Rightarrow \boxed{34.71 = P_5}$$

$D_4 = D_3(1+g)^n \Rightarrow 1(1+0.5)^1 = 1.5$

$D_5 = D_4(1+g)^n \Rightarrow 1.5(1+0.5)^1 = 2.25$

$D_6 = D_5(1+g)^n \Rightarrow 2.25(1+0.08)^1 = 2.43$

Calculate value of Stock at P_0:

$$P_0 = \frac{D_3}{(1+i)^n} + \frac{D_4}{(1+i)^n} + \frac{D_5}{(1+i)^n} + \frac{P_5}{(1+i)^n}$$

$$= \frac{1}{(1+0.15)^3} + \frac{1.5}{(1+0.15)^4} + \frac{2.25}{(1+0.15)^5} + \frac{34.71}{(1+0.15)^5}$$

$P_0 = 0.65 + 0.85 + 1.11 + 17.25$

$\boxed{P_0 = 19.86\$}$

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(7-13) $P = 100\$$

$D = 10\%$

$K_s = 8\% \rightarrow 0.08$

Annual Dividend = Stated Dividend rate * Dividend per value

Per share = $10\% * 100\$ = 10\$$.

(a) value of Preferred Stock At yeild 8%:

$K_s = D/P \Rightarrow P = D/K_s$

$P = 10/8\%$ $\boxed{P = 125\$}$

(b) Value of preferred stock at yeild of 12%.

$$P = \frac{D}{K_s} = \frac{10\$}{12\%} \Rightarrow \boxed{P = 83.33\$}$$

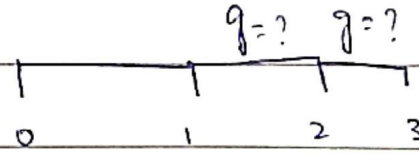
(7-14) $P_0 = 21.40\$$

$D_1 = 1.07$

$D_2 = 1.1449$

$D_3 = 1.2250$

$P_3 = 26.22$



Divided D_0.	1.07	1.1449	1.2250
			$P_3 = 26.22$

(a) Calculate growth Rate:

$D_2 = D_1(1+g)$

$g = \frac{D_2}{D_1} - 1$

$g = \frac{1.1449}{1.07} - 1$

$\boxed{g = 7\%}$

$D_3 = D_2(1+g)$

$g = \frac{D_3}{D_2} - 1$

$g = \frac{1.2250}{1.1449} - 1$

$\boxed{g = 6.99 \cong 7\%}$

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(b) Expected Dividend yeild:

Expected Dividend = $D_1/P_0 = \frac{1.07}{21.40} = 0.05$

(c) Total rate of Return:

Total rate of return = Expected Dividend + Capital Gain (growth)

$= 0.05 + 0.07 = 0.12$

$K_s = 12\%$

(7-15)

(a) $K_s = 15\% \rightarrow 0.15$

$D_0 = 2\$$, $g_a = -5\%$, $g_b = 0\%$, $g_c = 5\%$, $g_d = 10\%$

$P_0 = ?$

(b) $K_s = 15$

$g_a = 15\%$, $g_b = 20\%$

$P_0 = ?$ $P_0 = ?$

(a) Calculate the Present value of stock at different 'g' rate:

$g = -5\%$

$P_0 = \frac{D_0(1+g)}{K_s - g}$

$= \frac{2(1+(-0.05))}{0.15-(-0.05)}$

$= 1.9/0.2$

$P_0 = 9.5$

$g = 0\%$

$P_0 = \frac{D_0(1+g)}{K_s - g}$

$= \frac{2(1+0)}{0.15-0}$

$= 2/0.15$

$P_0 = 13.33$

$g = 5\%$

$P_0 = \frac{D_0(1+g)}{K_s - g}$

$= \frac{2(1+0.05)}{0.15-0.05}$

$= 2.1/0.1$

$P_0 = 21$

$g = 10\%$

$P_0 = \frac{D_0(1+g)}{K_s - g}$

$= \frac{2(1+0.10)}{0.15-0.10}$

$= 2.2/0.05$

$P_0 = 44$

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(b) Calculate value of stock (Gordan Constant Growth model) at growth rate 15% and 20%.

$g = 15\%$

$$P_0 = \frac{D_0(1+g)}{K_s - g}$$

$$P_0 = \frac{2(1+0.15)}{0.15 - 0.15}$$

$$\boxed{P_0 = 0}$$

$g = 20\%$

$$P_0 = \frac{D_0(1+g)}{K_s - g}$$

$$P_0 = \frac{2(1+0.20)}{0.15 - 0.20}$$

$$\boxed{P_0 = -48}$$

Explanation:

when required rate of return(K_s) equal the expected growth rate (g) the denominator of Gordan growth model became zero. This Indicate the stock value is infinite (Assum) bez dividend is expected to grow at same rate as required return. Lead to perpetual growth that model can't Compute.

when expected growth rate (g) is greater than required rate of return (K_s) the denominator ($K_s - g$) became negative. This result in -ve stock price which is unreasonable outcome. It Suggest market would not pay for stock if 'g' outpaces investor 'K_s'

(c) Is $g > K_s$ Reasonable: (growth rate is bigger the rate of return)

No, it's not reasonable to expect Constant growth ~~rate~~ stock would. have growth rate (g) greater than the required rate of return (K_s)

7-16 Data $R_{rf} = 11\%$

$R_m = 14\%$

$\beta = 1.5$

(a) $D_1 = 2.25\$$, $g = 5\%$ (constant)

CAPM → $K_s = R_{rf} + \beta (MRP)$

$= R_{rf} + \beta (R_m - R_{rp})$

$K_s = 11\% + 1.5 (14\% - 11\%)$

$K_s = 15.5\% \rightarrow 0.155$

$P_0 = \frac{D_1}{K_s - g} = \frac{2.25}{15.5\% - 5\%}$ = $21.42 ✓

(b) $R_{rf} = 9\%$
$R_m = 12\%$

$K_s = R_{rf} + \beta(MRP)$
$= R_{rf} + \beta(R_m - R_{rf})$
$= 9\% + 1.5(12\% - 9\%)$
$K_s = 0.135 = 13.5\%$

$P_0 = \frac{D_1}{K_s - g} = \frac{2.25}{13.5\% - 5\%}$ = $26.47 ✓

(c) $K_s = 9\% + 1.5(11\% - 9\%)$
$K_s = 0.12 = 12\%$

$P_0 = \frac{D_1}{K_s - g} = \frac{2.25}{12\% - 5\%}$ = $32.14 ✓

(d) $g = 6\%$, $\beta = 1.3$, $R_{rf} = 9\%$, $R_m = 11\%$
$K_s = R_{rf} + \beta(MRP)$
$K_s = R_{rf} + \beta(R_m - R_{rf})$
$K_s = 9\% + 1.3(11\% - 9\%)$
$K_s = 0.116 \rightarrow 11.6\%$

$P_0 = \frac{D_1}{K_s - g} = \frac{2.25}{11.6\% - 6\%}$ = $40.17 ✓